# Fixed-Stitch Ref–GH Failure Evidence and a Controlled Gaussian-Reference / Full-$\gamma_2$ Investigation

AthenaK vacuum Ref–GH campaign report

24 August 2026

**Abstract**

This report packages the compact evidence from the paused Aurora fixed-core Schwarzschild continuation campaign and turns its formulation audit into a controlled next-step design. The feedback run T4 failed closed at $t = 3.70126M$; the independent prescribed run T5 became ill-conditioned at RK stage time $t = 3.18677M$. Growth is localized near $r = 0.45$–$0.56M$, within the artificial $0.30$–$0.60M$ wormhole/trumpet stitch, while measured characteristic travel excludes the $24M$ outer boundary. Feedback delays failure in coordinate time but is worse than the prescribed path at equal activation near the transition. The present code has $\gamma_0 > 0$ but $\gamma_2 = 0$; its "standard" $\Phi$ ordering is not the complete $\gamma_2$-damped first-order generalized-harmonic system.

The proposed follow-up separates two hypotheses: replace the narrow compact stitch with a broad Gaussian-localized Schwarzschild reference, then add the complete standard $\gamma_2$ constraint additions in the moving reference frame. No Gaussian or nonzero-$\gamma_2$ evolution has yet been run. This document therefore establishes neither trumpet stability nor convergence; it defines the derivation, oracles, fixed matrices, and fail-closed gates required to test those claims.

## 1 Status and claim boundary

The report branch was created only after fetching the remote parent and confirming exact equality:

| | |
|---|---|
| parent branch | `codex/ref-gh-feedback-continuation-20260823` |
| parent/local/upstream/remote SHA | `e5579269f2e979d246ab162a288b140e7076d666` |
| report branch | `codex/ref-gh-gaussian-reference-gamma2-20260824` |
| T5 source SHA | `0189d495058e09a53ca36a31f08ab924bc496582` |
| T5 executable SHA-256 | `7496ae64fe8a57c4f3111884bd04be94`... |

The complete executable hash and run provenance are retained in the compact artifact tree. The final pushed branch SHA is deliberately reported by Git rather than embedded self-referentially in this PDF. A live `qstat -u hzhu` check during packaging returned no entries, confirming that no Aurora campaign job remained queued or running.

**Established.** The T0–T2 source/algebra/moving-reference gates, a twelve-PVC execution gate, rank-to-tile mapping, and checkpoint/restart continuity passed for their named workloads. T4 and T5 both failed on the same 328-MeshBlock, $[-24M, 24M]^3$ medium SMR mesh. Their failure maxima are deep inside the finest region and causally disconnected from the outer faces.

**Not established.** No result supports stable full activation, a stationary trumpet hold, resolution convergence, long-time stability, production performance, or a causal attribution to one formulation term. No Gaussian reference, GH-compatible mismatched-reference initializer, or $\gamma_2$ implementation exists on this report branch. All proposed matrices in Sec. 6 therefore remain not run.

## 2 Controlling fixed-stitch evidence

### 2.1 Runs and outcomes

Both discriminators used eight Aurora nodes, 96 distinct PVC tiles, one rank per tile, and the identical physical grid. The minimum spacing was $\Delta x_{\min} = M/24$. No floor, clipping, stuffing, reset, or threshold relaxation concealed either failure.

Table 1: Definitive medium-grid outcomes. History output precedes the fatal RK-stage check.

| run | last history $t/M$ | fatal stage $t/M$ | last $\xi$ | $S(\xi)$ | outcome |
|---|---|---|---|---|---|
| T4 feedback | 3.652312 | 3.70126 | 0.683571 | 0.814516 | constraint-growth veto |
| T5 prescribed | 3.150078 | 3.18677 | 0.787520 | 0.932046 | invalid relative metric |

Figure 1 shows the complete committed histories. T5 traverses the reference path faster and reaches substantially larger conditioning, relative boost, and native constraints by its final output. This is a time-domain observation, not evidence that feedback repairs the formulation.

### 2.2 Equal-activation comparison

At a common coordinate time, feedback has activated less of the reference and therefore looks much better. The scientifically relevant discriminator is equal $\xi$. Figure 2 shows that the histories are close through $\xi \simeq 0.60$, after which the controlled path is worse. At $\xi = 0.675$, T4 has condition number 6.4819 versus 3.5563, relative-boost diagnostic 1.7117 versus 0.9850, GH norm $1.4082\times10^{-2}$ versus $4.8002\times10^{-3}$, reduction norm $1.6242 \times 10^{-3}$ versus $5.1338 \times 10^{-4}$, and curl norm $9.0562 \times 10^{-2}$ versus $3.1148 \times 10^{-2}$. Feedback delayed the event in coordinate time but did not improve the state at equal activation.

### 2.3 Localization and causal exclusion

The T4 accumulated characteristic distance is $3.48176M$, leaving $20.51824M$ to the nearest outer face. T5 accumulated $3.00307M$, leaving $20.99693M$. Thus neither outer boundary nor restart discontinuity can explain these events. Figure 3 instead places the maxima of the GH, reduction, curl, and reference-frame source diagnostics in the artificial 0.30–0.60$M$ stitching shell, well inside the finest-box face at $r = 4M$. This localization motivates the Gaussian hypothesis but does not prove it.

## 3 Current formulation and the missing standard $\gamma_2$ system

Let $E_I = E_I{}^i\partial_i$ be the reference spatial frame, with dual $\theta^I{}_i$ and commutator $[E_I, E_J] = f_{IJ}{}^K E_K$. The evolved variables are the frame metric $\Psi_{AB}$, its normal-time variable $\Pi_{AB}$, and spatial derivative $\Phi_{IAB}$. The primary first-order constraints are

$$\mathcal{C}_{IAB} = E_I(\Psi_{AB}) - \Phi_{IAB}, \tag{1}$$

$$\mathcal{C}_{IJAB} = E_I(\Phi_{JAB}) - E_J(\Phi_{IAB}) - f_{IJ}{}^K\Phi_{KAB}. \tag{2}$$

The second identity is the noncoordinate exterior derivative and is essential: omitting $f_{IJ}{}^K$ creates a false curl signal.

The implemented system has effective $\gamma_1 = -1$, gauge damping $\gamma_0 > 0$, and $\gamma_2 = 0$. Its compatible update differentiates the discrete $\Psi$ right-hand side and adds $(\partial_t E_I{}^p)\theta^J{}_p\Phi_J$. The optional standard ordering subtracts $\beta^J\mathcal{C}_{IJAB}$; this changes a curl-equivalent principal ordering but does not supply standard reduction-constraint damping. Component-wise KO dissipation and component-wise SMR transfer also need not preserve $\Phi = E(\Psi)$ in a varying frame.

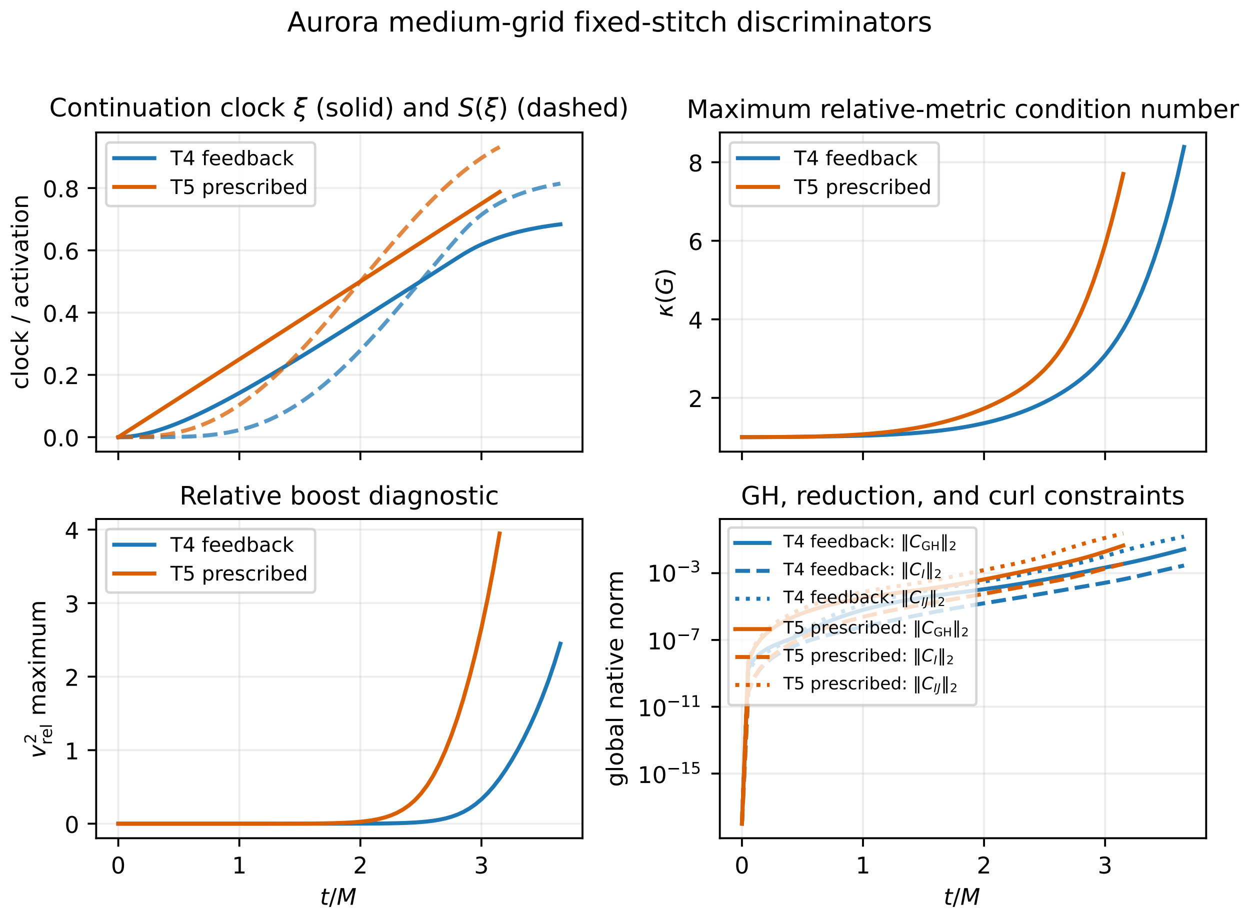


Figure 1: T4 feedback and T5 prescribed fixed-stitch histories. Solid/dashed clock curves distinguish $\xi$ from the quintic activation $S(\xi)$. The constraints are native global $L_2$ diagnostics.

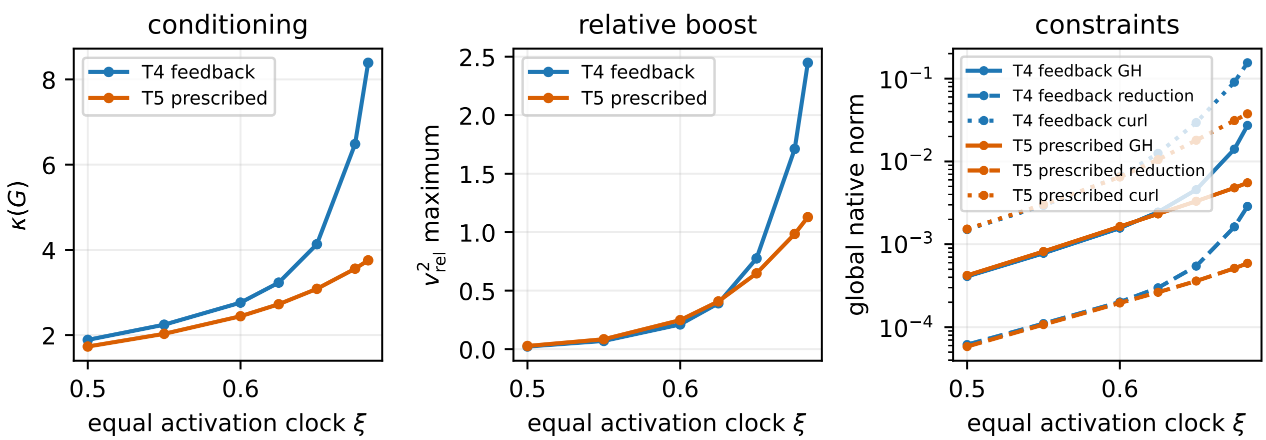


Figure 2: Comparison at identical continuation clock $\xi$. The controlled path is not superior near the failure transition.

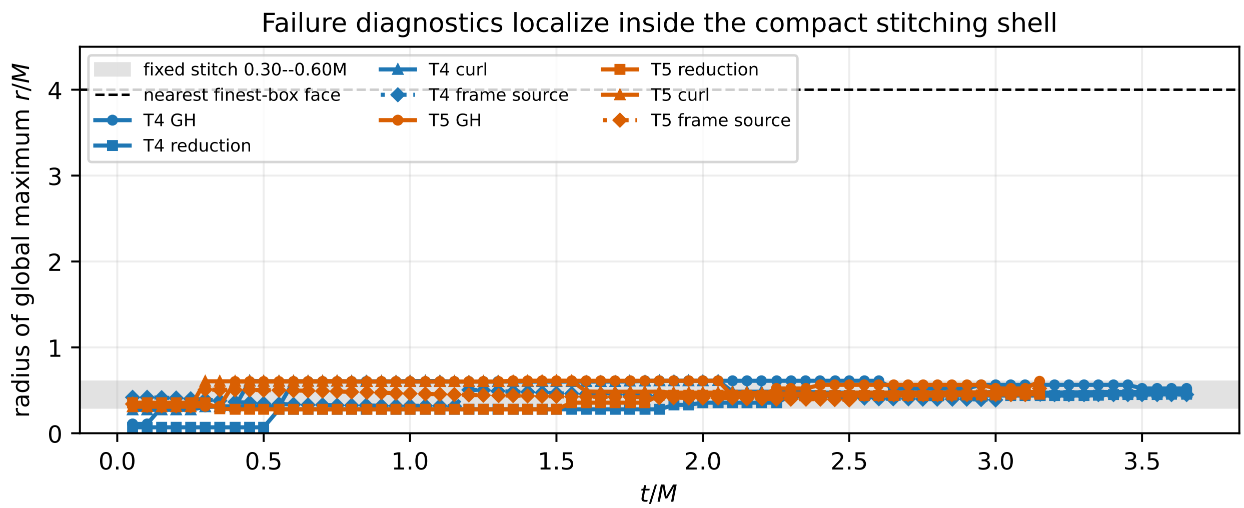


Figure 3: Radii of global diagnostic maxima. Duplicate checkpoint rows are deduplicated by time and diagnostic. The refinement interface is not the observed maximum location.

### 3.1 Coordinate standard system

The complete first-order generalized-harmonic system of Lindblom, Scheel, Kidder, Owen, and Rinne adds matched multiples of the reduction constraint to all required equations. Suppressing non-$\gamma_2$ sources, its principal form is

$$\partial_t\psi_{ab} - (1+\gamma_1)\beta^k\partial_k\psi_{ab} = -\alpha\Pi_{ab} - \gamma_1\beta^i\Phi_{iab}, \tag{3}$$

$$\partial_t\Pi_{ab} - \beta^k\partial_k\Pi_{ab} + \alpha g^{ki}\partial_k\Phi_{iab} - \gamma_1\gamma_2\beta^k\partial_k\psi_{ab} = \cdots - \gamma_1\gamma_2\beta^i\Phi_{iab}, \tag{4}$$

$$\partial_t\Phi_{iab} - \beta^k\partial_k\Phi_{iab} + \alpha\partial_i\Pi_{ab} - \alpha\gamma_2\partial_i\psi_{ab} = \cdots - \alpha\gamma_2\Phi_{iab}. \tag{5}$$

Symmetric hyperbolicity fixes $\gamma_3 = \gamma_1\gamma_2$; this is why one guessed local term is not an admissible comparator.

For $\gamma_1 = -1$, the additional right-hand sides can be written directly in constraint form:

$$\delta_{\gamma_2}(\partial_t \Psi_{AB}) = 0, \tag{6}$$

$$\delta_{\gamma_2}(\partial_t \Pi_{AB}) = -\gamma_2 \beta^I \mathcal{C}_{IAB}, \tag{7}$$

$$\delta_{\gamma_2}(\partial_t \Phi_{IAB}) = +\alpha\gamma_2 \mathcal{C}_{IAB}. \tag{8}$$

Equation (8) is the compact implementation oracle, not a license to omit the matching characteristic variables, diagnostics, and boundary implications.

### 3.2 Moving-frame and subsidiary requirements

Because Eq. (8) is algebraic in the geometrically defined constraint, it transforms cleanly to the reference frame. All existing time-dependent-frame and structure-coefficient terms remain part of the core system. If

$$T_I{}^J = (\partial_t E_I{}^p)\theta^J{}_p, \tag{9}$$

then the reduction subsidiary equation must include the corresponding $T_I{}^J \mathcal{C}_J$ coupling, while the curl equation must include the induced two-form action and the time derivative of the structure coefficients. These are homogeneous constraint couplings; they cannot be dropped merely because they vanish on the constraint surface.

The cited standard system gives the principal reduction-constraint speed $(1+\gamma_1)\beta^n$. Therefore, with $\gamma_1 = -1$, its frozen-coefficient subsidiary equation is locally

$$\partial_t \mathcal{C}_{IAB} \simeq -\alpha\gamma_2 \mathcal{C}_{IAB}, \tag{10}$$

not literally $(\partial_t - \beta^I E_I)\mathcal{C}_{IAB}$ unless a different convention or constraint addition is deliberately chosen. This is an important correction to the schematic target in the controlling specification and must be resolved by a symbolic principal-symbol oracle before production code is edited.

Taking the noncoordinate exterior derivative of Eq. (10) gives, schematically,

$$\partial_t \mathcal{C}_{IJAB} \simeq -\alpha\gamma_2 \mathcal{C}_{IJAB} - 2E_{[I}(\alpha\gamma_2)\mathcal{C}_{J]AB} + \text{homogeneous frame/shift couplings.} \tag{11}$$

Thus the physically relevant curl is damped in the frozen-coefficient limit; an independent direct curl penalty must not be invented without a separate hyperbolicity proof.

The characteristic fields become

$$u^{\hat{0}}_{AB} = \Psi_{AB}, \tag{12}$$

$$u^{\hat{1}\pm}_{AB} = \Pi_{AB} \pm n^I \Phi_{IAB} - \gamma_2 \Psi_{AB}, \tag{13}$$

$$u^{\hat{2}}_{IAB} = (\delta_I{}^J - n_I n^J)\Phi_{JAB}, \tag{14}$$

with speeds $-(1+\gamma_1)\beta^n$, $-\beta^n \pm \alpha$, and $-\beta^n$, respectively. The transformed symmetrizer contains

$$\Lambda^2\, d\Psi^2 + d\Pi^2 - 2\gamma_2\, d\Psi\, d\Pi + \bar{g}^{IJ} d\Phi_I d\Phi_J, \qquad \Lambda^2 > \gamma_2^2. \tag{15}$$

These formulas define unit tests for signs, characteristic reconstruction, and positive definiteness.

## 4 Gaussian-localized Schwarzschild reference

The proposed reference replaces the sole narrow spatial transition by

$$W(r; R_G) = \exp[-(r/R_G)^2], \qquad \rho = r/M, \tag{16}$$

without compact support. The time homotopy is $s(t) = S(t/\tau_{\text{ramp}})$, where, for $0 < u < 1$,

$$S(u) = 10u^3 - 15u^4 + 6u^5, \tag{17}$$
$$S'(u) = 30u^2(1-u)^2, \tag{18}$$
$$S''(u) = 60u(1-u)(1-2u). \tag{19}$$

The implementation must branch at the endpoints so that $\dot{s} = \ddot{s} = 0$ exactly for $t \geq \tau_{\text{ramp}}$ and before the ramp.

The unlocalized black-hole primitives are

$$\log \alpha_{\text{BH}} = (1-s)\log \alpha_W + s \log \alpha_T, \tag{20}$$
$$\log \psi^2_{\text{BH}} = (1-s)\log \psi^2_W + s \log \psi^2_T, \tag{21}$$
$$q^\beta_{\text{BH}} = s q^\beta_T, \tag{22}$$

and Gaussian localization relative to Minkowski is

$$\log \bar{\alpha} = W \log \alpha_{\text{BH}}, \tag{23}$$
$$\log \bar{\psi}^2 = W \log \psi^2_{\text{BH}}, \tag{24}$$
$$\bar{q}^\beta = W q^\beta_{\text{BH}}, \qquad \bar{\beta}^i = \bar{q}^\beta x^i. \tag{25}$$

It approaches the local hole reference at the puncture and Minkowski smoothly for $r \gg R_G$. The analytic radial jets are

$$W' = -\frac{2r}{R_G^2} W, \tag{26}$$
$$W'' = \left(\frac{4r^2}{R_G^4} - \frac{2}{R_G^2}\right) W, \tag{27}$$

while $\dot{s} = S'(u)/\tau_{\text{ramp}}$ and $\ddot{s} = S''(u)/\tau^2_{\text{ramp}}$. Products, logarithms, and exponentials must use the existing analytic `ReferenceJet` algebra; finite differencing the reference is outside scope.

Figure 4 illustrates the predeclared widths. It is a design plot only. The broad derivative scales are controlled by $R_G^{-1}$ and $R_G^{-2}$, which the reference-only scan must distinguish from logarithmic or algebraic grid-scale growth.

## 5 GH-compatible initialization with a mismatched reference

The physical initial data remain exact isotropic Schwarzschild wormhole data:

$$\gamma_{ij} = \psi_W^4 \delta_{ij}, \quad \alpha = \psi_W^{-2}, \quad \beta^i = 0, \quad K_{ij} = 0. \tag{28}$$

Because the Gaussian reference becomes Minkowski outside its local region, $\Psi = \eta$, $\Pi = 0$, and $\Phi = 0$ are generally false. A reusable ADM-plus-reference initializer should proceed as follows.

1. Construct the physical coordinate four-metric,
$$g_{00} = -\alpha^2 + \gamma_{ij}\beta^i\beta^j, \qquad g_{0i} = \gamma_{ij}\beta^j, \qquad g_{ij} = \gamma_{ij}. \tag{29}$$
2. Transform with the analytic reference frame: $\Psi_{AB} = e_A{}^a e_B{}^b g_{ab}$.
3. Differentiate the analytic physical metric and reference frame, then set $\Phi_{IAB} = E_I{}^i \partial_i \Psi_{AB}$. This step includes derivatives of both $g_{ab}$ and $e_A{}^a$.

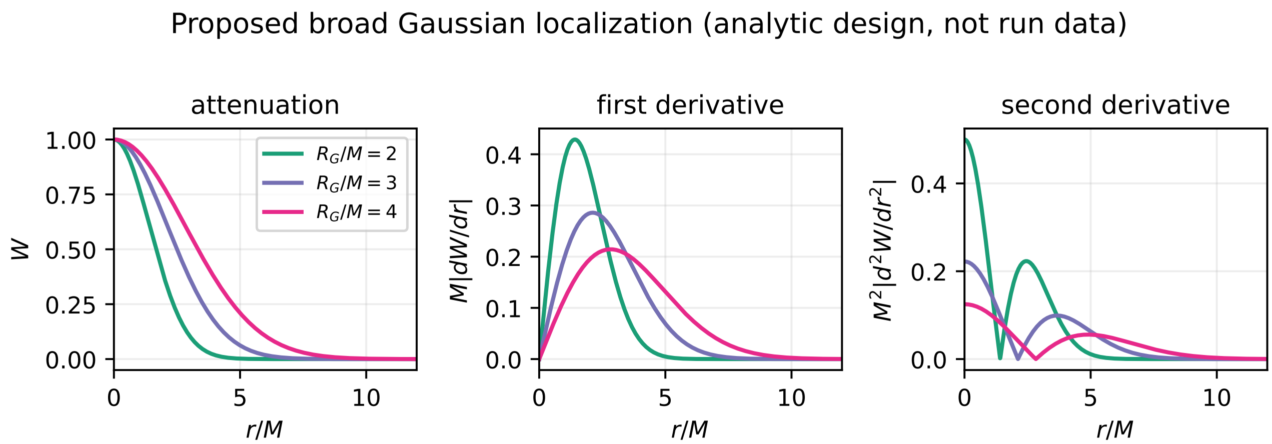


Figure 4: Analytic Gaussian attenuation and derivative magnitudes for the fixed proposed widths. No evolution data enter this figure.

4. Use $\partial_t \gamma_{ij} = -2\alpha K_{ij} + \mathcal{L}_\beta \gamma_{ij}$ to fix the six spatial-metric time derivatives. Determine the remaining four $\partial_t g_{0\mu}$ by solving the four background-covariant GH gauge constraints $\mathcal{C}_A = 0$. Equivalently, solve the resulting ten-by-ten linear map for $\Pi_{AB}$ after including the analytic $\partial_t {e_A}^a$ terms. Do not guess $\Pi$.

5. Reconstruct ADM variables and require the intended $(\alpha, \beta^i, \gamma_{ij}, K_{ij})$, roundoff-level analytic $\Phi$ definition residual, roundoff/truncation-oracle GH constraint, and existing Schwarzschild ADM-constraint accuracy.

This linear solve is the correct place to expose rank, conditioning, and sign failures. It is also the reusable architecture needed by a future composite binary reference; a generic Cholesky tetrad is deliberately not part of the present task.

## 6 Predeclared gates and fixed experiments

The Gaussian and $\gamma_2$ hypotheses must be separated before being combined. Table 2 records the current evidence boundary.

Table 2: Completion state at report generation.

| gate | state | evidence |
|---|---|---|
| existing source/moving-reference/PVC/restart gates | passed | committed logs |
| T4 medium feedback continuation | failed | job 8777607 |
| T5 medium prescribed continuation | failed | job 8777824 |
| Gaussian reference and analytic jet oracle | not run | no implementation |
| GH-compatible mismatched-reference initializer | not run | no implementation |
| standard moving-frame $\gamma_2$ manufactured tests | not run | no implementation |
| Gaussian $3 \times 3$ reference-only scan | not run | no results |
| Gaussian $3 \times 3$ medium evolution matrix | not run | no results |
| Gaussian/$\gamma_2$ black-hole ablation and causal $2 \times 2$ | not run | no results |
| conditional three-resolution gate | disallowed now | prerequisite unmet |

### 6.1 Reference-only matrix

Before evolution, scan exactly $R_G/M \in \{2,3,4\}$ and $\tau_{\rm ramp}/M \in \{4,8,16\}$ at $s \in \{0, 0.25, 0.5, 0.75, 1\}$ and cell-centered radii for $h/M \in \{1/16, 1/24, 1/32, 1/48\}$. Record values and locations of $|W'|$, $|W''|$, spin connection and derivative, reference Ricci/Riemann, every source sector, and the total source. Compare the maxima with the old compact stitch and fit the innermost resolution dependence without expanding the matrix after seeing results.

### 6.2 Manufactured off-constraint tests

Plant reduction and curl violations separately in flat, spatially varying, and genuinely time-dependent frames for $\gamma_2 M = 0, 0.5, 1.0$. The frozen-coefficient reduction norm must follow the decay rate derived from Eq. (10); curl behavior must follow the exterior-derivative subsidiary system. Repeat selected cases with KO off/on and record an RHS split into core, moving-frame, $\gamma_2$, and KO contributions. A zero-constraint exact solution is not a sufficient test.

### 6.3 Black-hole matrices

First run the fixed Gaussian $3 \times 3$ medium matrix with standard ordering, $\gamma_2 = 0$, controller off, and $\delta_q = \delta_p = 0$. Rank cases lexicographically by: successful full activation plus hold, then maximum relative boost, condition number, GH/curl growth, and reference/source maxima. Promote at most two candidates. Only those candidates receive $\gamma_2 M = 0, 0.5, 1.0$; 2.0 is allowed once only under the predeclared narrow condition. Finally compare old/Gaussian crossed with $\gamma_2 M = 0/1$ at equal time and activation.

The three-resolution $M/16$, $M/24$, $M/32$ gate is conditional on one medium candidate reaching full activation plus a $2M$ hold with controlled native constraints. Until then there is no convergence campaign to report.

## 7 Portability, SMR, and boundary implications

Aurora qualification must show Level Zero SYCL execution on each requested PVC tile, not merely configuration or compilation. Runtime evidence must retain source and Kokkos SHAs, compiler/runtime/device versions, selectors, rank-to-tile mapping, binary hashes, checkpoints, and resource observations. The current T4/T5 evidence satisfies this for those exact runs only.

Two discrete constraint-injection channels require instrumentation: component-wise KO does not commute with $\Phi = E(\Psi)$ in a spatially varying frame, and AthenaK transfers all 50 evolved components independently across SMR. Neither mechanism is established as the cause of T4/T5 because maxima lie near $r = 0.5M$, far inside the nearest interface. If a manufactured or Gaussian run localizes first growth at an interface, the black-hole gate must stop for a separate constraint-compatible transfer/gradient-repair task.

The existing outer treatment is not a constraint-preserving GH boundary. Although causally irrelevant to T4/T5, it remains a future long-time qualification blocker. Every subsequent run must integrate maximum characteristic speed and state the remaining coordinate distance rather than calling the boundary remote by inspection.

## 8 Reproduction and artifact map

From the repository root, regenerate all figures and compile the report with:

```
python3 scripts/ref_gh/plot_ref_gh_gaussian_gamma2_report.py
```

```
cd docs
pdflatex -interaction=nonstopmode -halt-on-error \
  ref_gh_gaussian_reference_gamma2_20260824.tex
pdflatex -interaction=nonstopmode -halt-on-error \
  ref_gh_gaussian_reference_gamma2_20260824.tex
```

The script reads only committed T4/T5 histories and the committed equal-clock comparison JSON. The generated plot PDFs/PNGs and this report PDF live under `docs/fo_gh_artifacts/ref_gh_gaussian_reference_gamma2_20260824/` or alongside this source, as recorded in the artifact manifest.

The large Aurora outputs remain intentionally outside Git:

- T4: `/lus/flare/projects/CompactBinaryMerger/hzhu/refgh_feedback_continuation_20260823_2466879e_v1/runs/t4_feedback_outer24_8777607.aurora-pbs-0001.hostmgmt.cm.aurora.alcf.anl.gov`
- T5: `/lus/flare/projects/CompactBinaryMerger/hzhu/refgh_feedback_continuation_20260823_2466879e_v1/runs/t5_prescribed_outer24_8777824.aurora-pbs-0001.hostmgmt.cm.aurora.alcf.anl.gov`

# 9 Conclusion

The compact fixed-core evidence rejects both tested continuation paths but does not identify a unique cause. The strongest observations are the stitch-shell localization and the absence of standard $\gamma_2$ damping; the strongest structural risks are moving-frame off-constraint propagation, component-wise KO, and independent SMR transfer. The Gaussian and $\gamma_2$ hypotheses must therefore be tested independently with analytic jets, a GH-compatible initializer, planted off-constraint oracles, and fixed predeclared matrices. At this point the correct scientific statement is:

> No Gaussian-reference result, no $\gamma_2$ damping result, no convergent trumpet evolution, and no long-time stability result has been established.

# References


[1] L. Lindblom, M. A. Scheel, L. E. Kidder, R. Owen, and O. Rinne, "A New Generalized Harmonic Evolution System," Class. Quantum Grav. 23 (2006), https://arxiv.org/abs/gr-qc/0512093.